

From Philosophy to XAI (via Psychology)



Ulrike Kuhl

(slides created in cooperation with Mark T. Keane, UCD)

Counterfactual Perspectives

Reality: not X, and Y happened

Counterfactual: If only X, then Y would have differed

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- How do we handle them in a truth-functional way?
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Philosophy :

- How do we handle them in a truth-functional way?
- Counterfactuals and scientific causality

Psychology:

- Thinking about the past, future, causes (+ distinct emotions)
- What gets changed? Mutability
- Theories of counterfactual thinking...

PHILOSOPHY

The Best of All Possible Worlds...





Logical Issues: Truth



"If kangaroos had no tails, then they would topple over"



Logical Issues: Truth



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- Counterfactuals can be true or false but establishing that truth is non-trivial (Nickerson, 2015)



Logical Issues: Truth



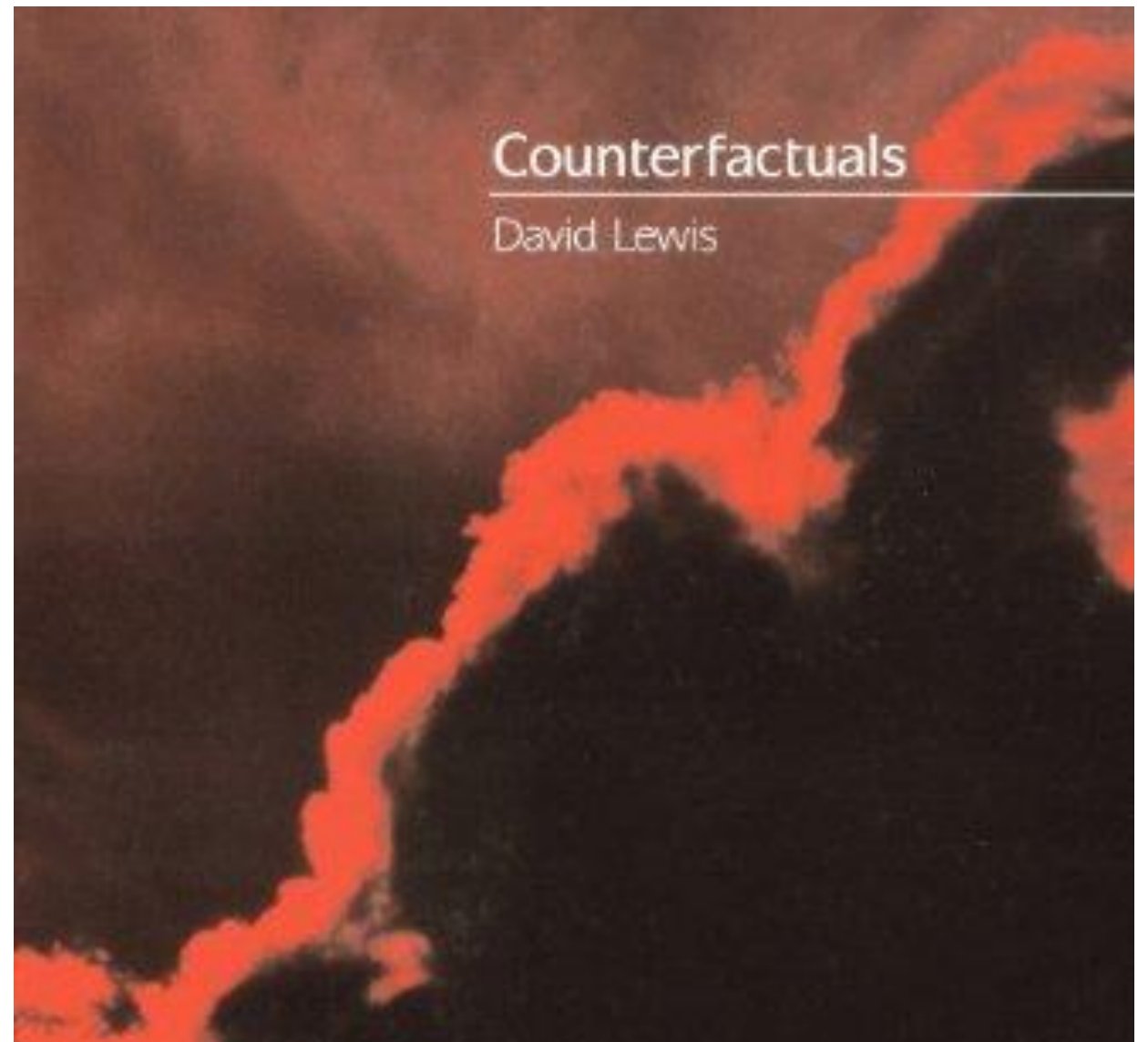
"If kangaroos had no tails, **then** they would topple over"

- Counterfactuals can be true or false but establishing that truth is non-trivial (Nickerson, 2015)
- **Problem**: The ***if-part*** is false and the ***then-part*** is false, so truth cannot be function of its components (and yet?)
- **Solution**: Counterfactual is true in a *possible world*, that is otherwise the same as this world (Stalnaker, 1968)

Counterfactual reasoning considers the closest relevant possible world(s) in which the antecedent holds...



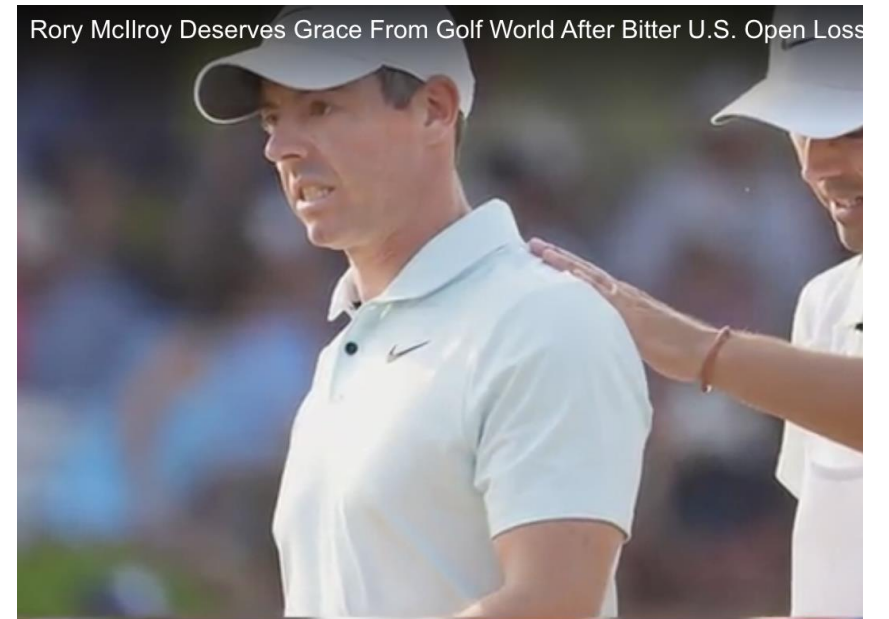
David Kellogg Lewis (1941-2001)



Lewis (1973) *Counterfactuals*, CUP.

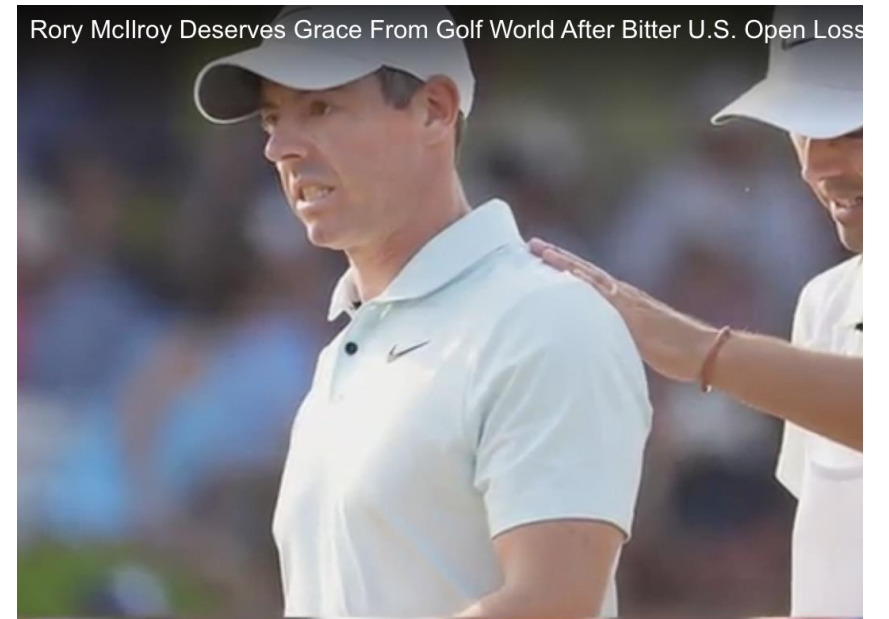
Recommended reading: [Lewis, D. \(2013\). *Counterfactuals*. John Wiley & Sons.](#)

***"Minimal Changes" &
"Closeness"
seem slippery...***

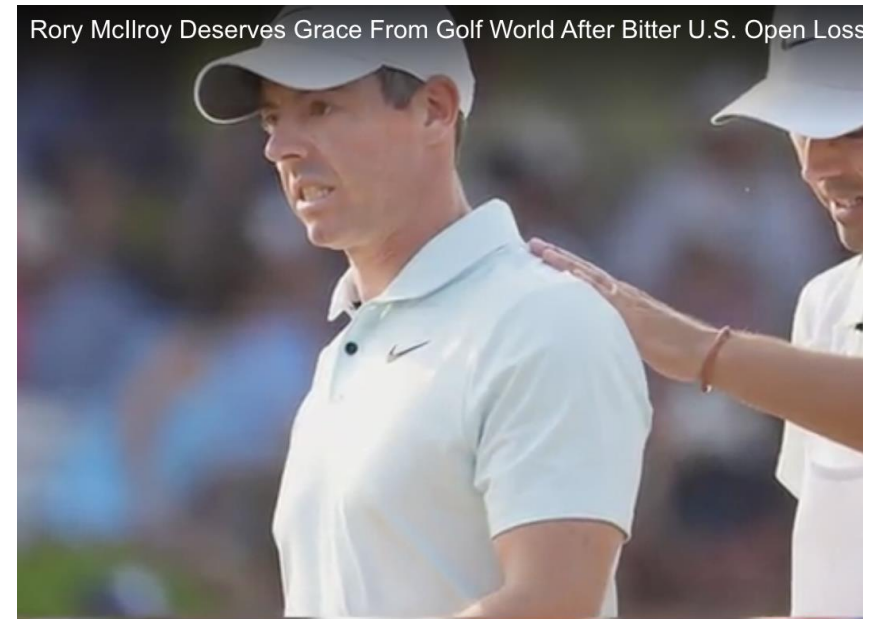


"Minimal Changes" & "Closeness" seem *slippery*...

- What's a minimal change?

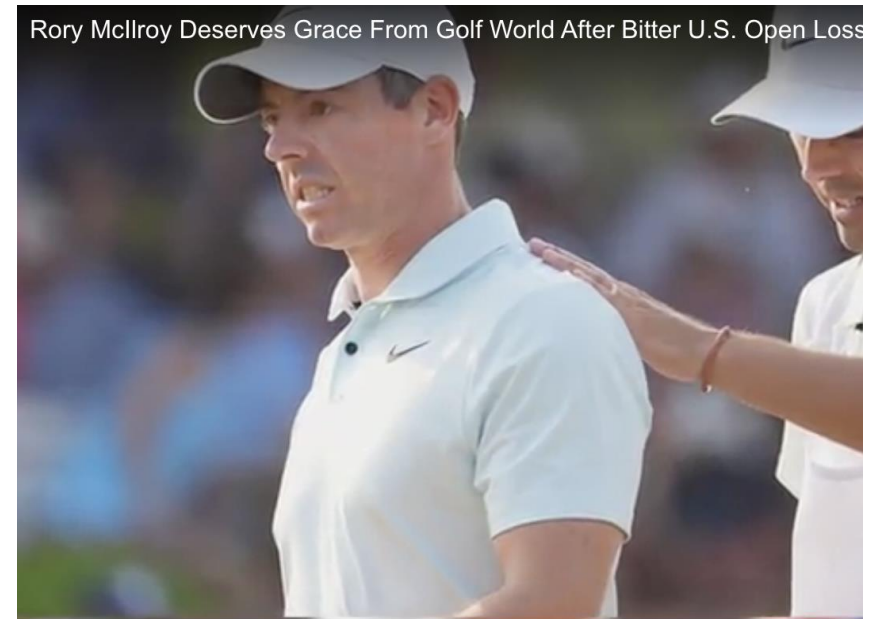


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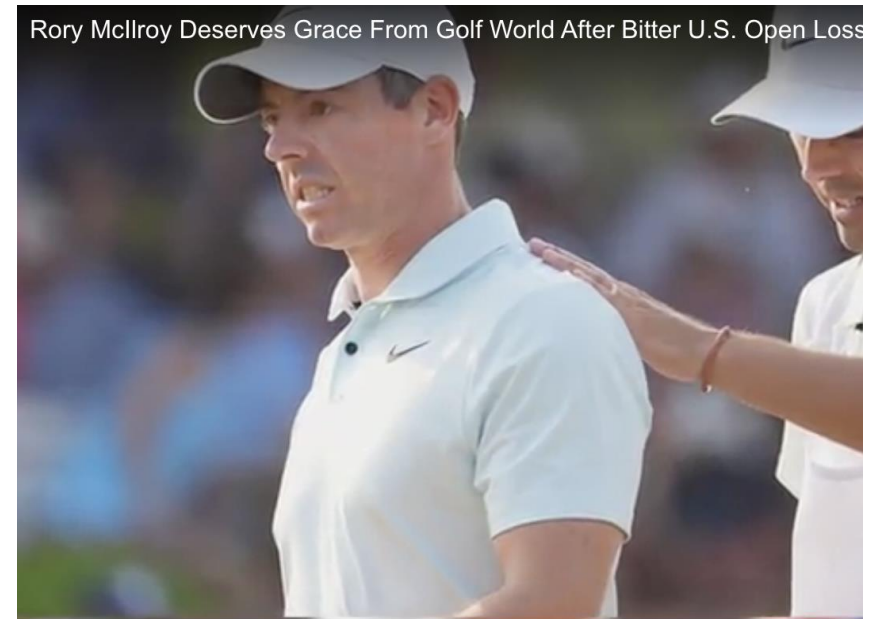
- What's a minimal change?
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"Minimal Changes" & "Closeness" seem slippery...



- What's a minimal change?
- What's close will depend (on the context, on the individual...)
- Possible worlds get complex (what is true in all possible worlds?)
- Different logics have been proposed to handle issues

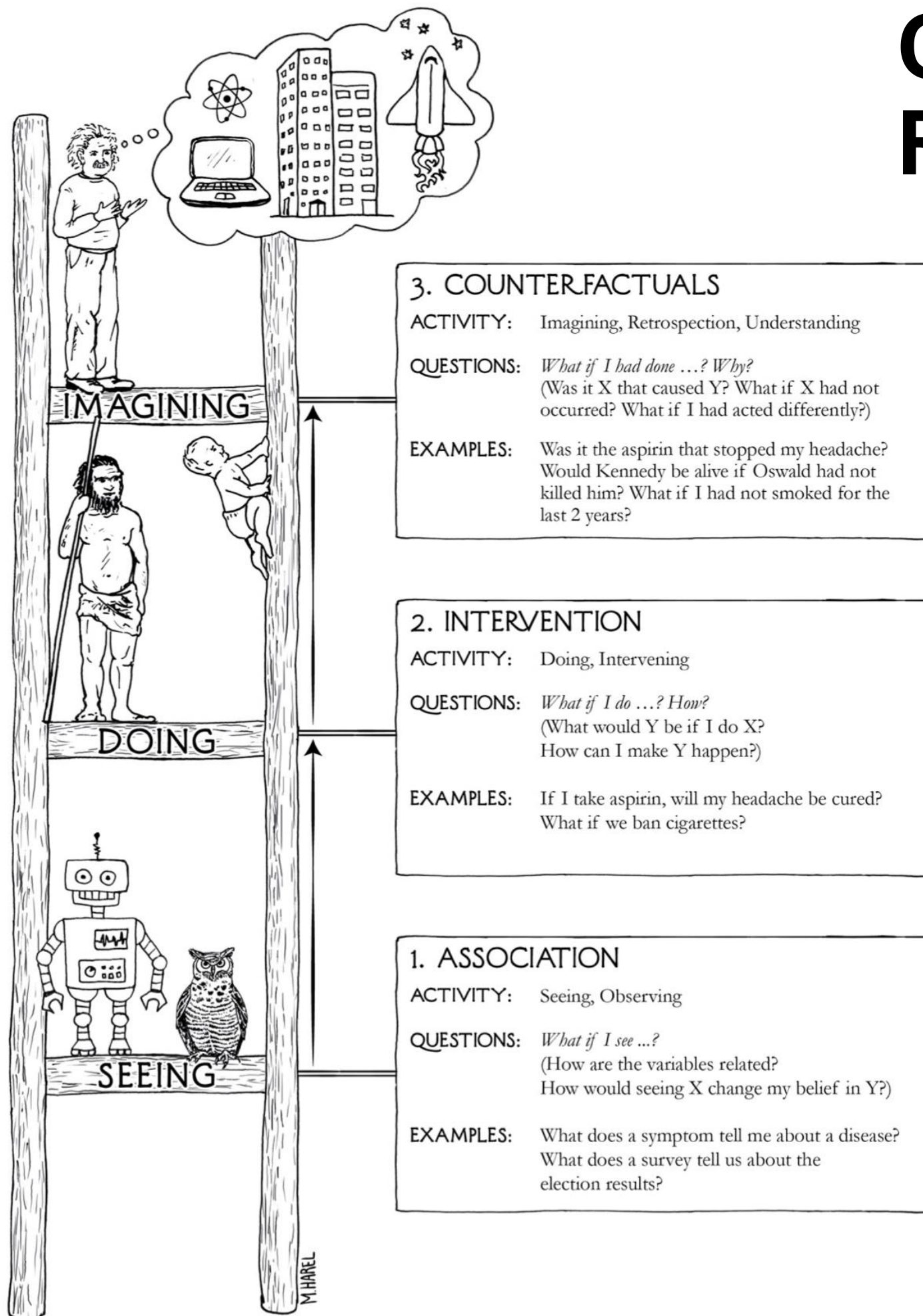
Counterfactuals and Roots of Causality

- Woodward's (2005) interventionist account of causation in Philosophy of Science:

A causal relationship between X and Y exists if, under an ideal intervention (all other variables are controlled/held-constant), a change in X leads to a change in Y

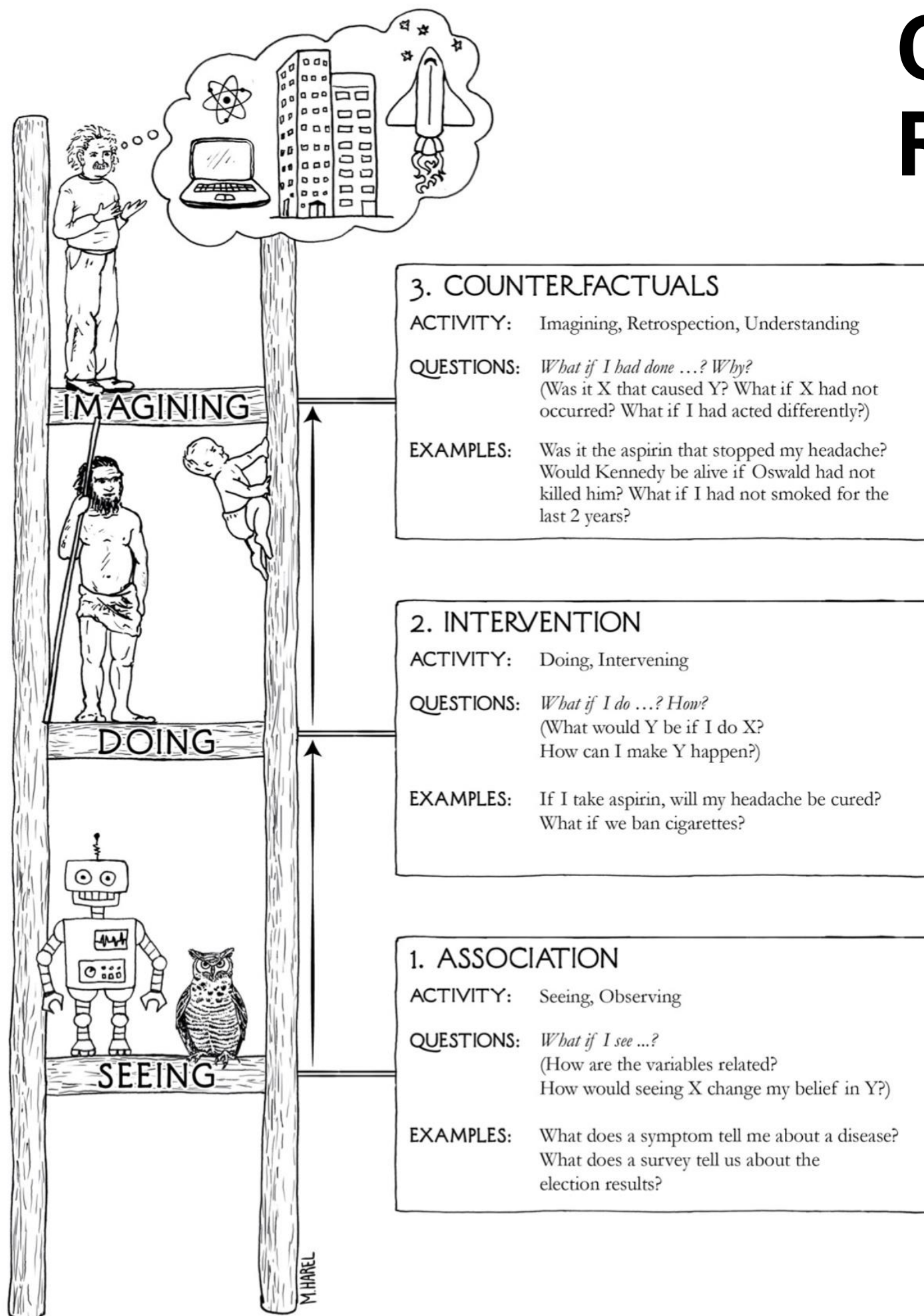
→ "If X had not occurred, Y would not have occurred."

Counterfactuals and Roots of Causality



Drawing by Maayan Harel, The Ladder of Causation, as printed in [Pearl, Judea; Mackenzie, Dana \(2020\) "The Book of Why: The new Science of Cause and Effect"](#), Basic Books, Copyright © 2018 by Judea Pearl and Dana Mackenzie

Counterfactuals and Roots of Causality



„Counterfactual learners can imagine worlds that do not exist and infer reasons for observed phenomena“

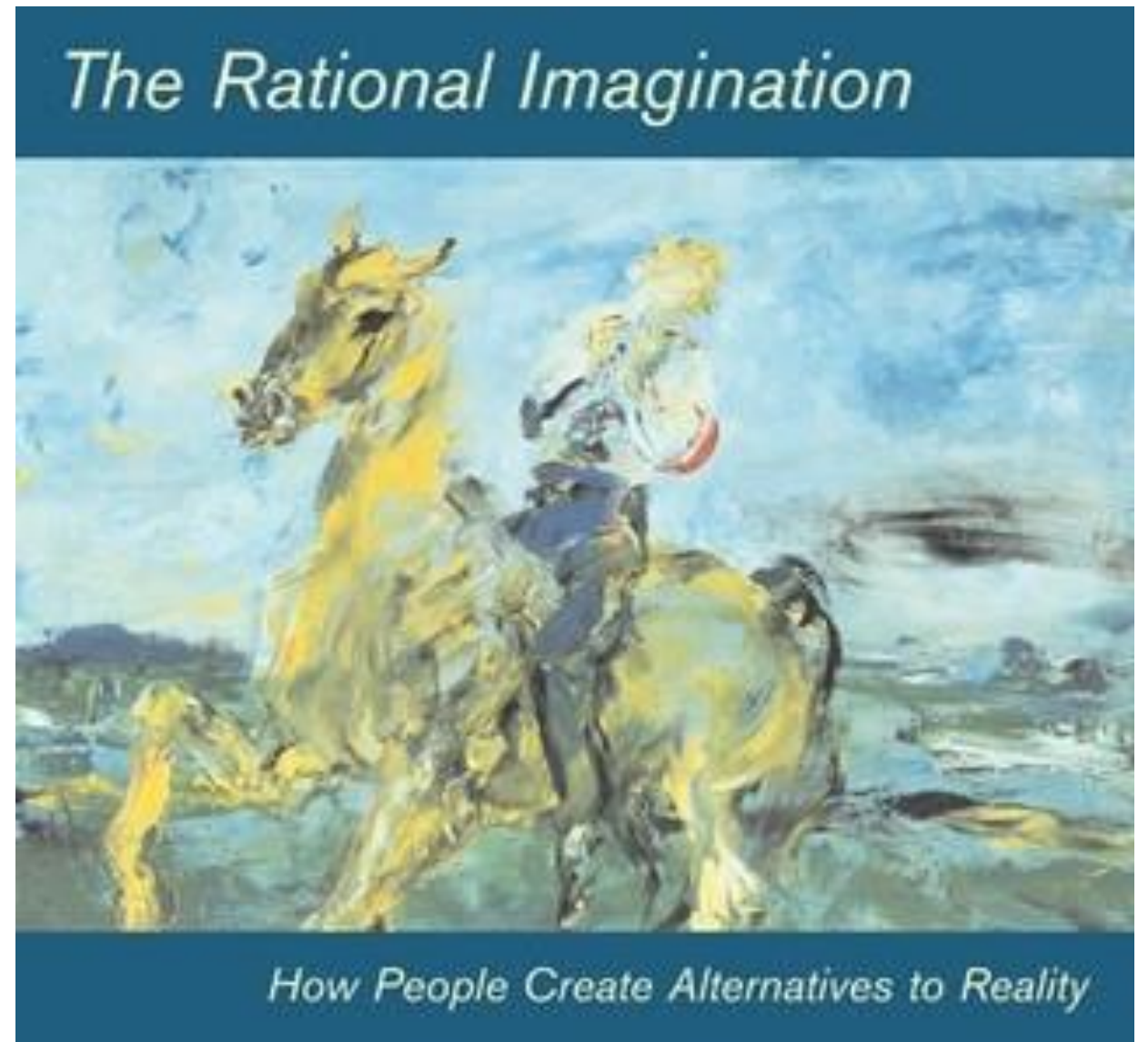
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PSYCHOLOGY

The Fault Lines of Reality...



Daniel Kahneman (1934-2024)



[Byrne \(2007\) MIT Press](#)

Counterfactual Thinking

- Critical to thinking about our past, future, causality, moral judgements (blame) with emotional importance (regret, guilt, blame)
 - ~ “If only I hadn’t married that idiot, I would be happy.”
 - ~ “I’ll do better in my next exam, if I concentrate more.”

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 - ~ “If only I hadn’t married that idiot, I would be happy.”
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- Thoughts about how things could have turned out differently can lead to *regret* (influencing decisions), *guilt* and *self-blame*

Fault Lines of Reality: What to Change?

Sam left work at the usual time, taking a different route home and at a junction near his house, a drunk trucker swerved into his car, killing him outright.

If only Sam had... what?

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If only Sam had... what?

- ... taken his usual route home
- ... not left at his usual time
- ... taken his bike home, instead
- ... not gone to work that day
- ... had faster reactions
- ... the truck had not swerved !

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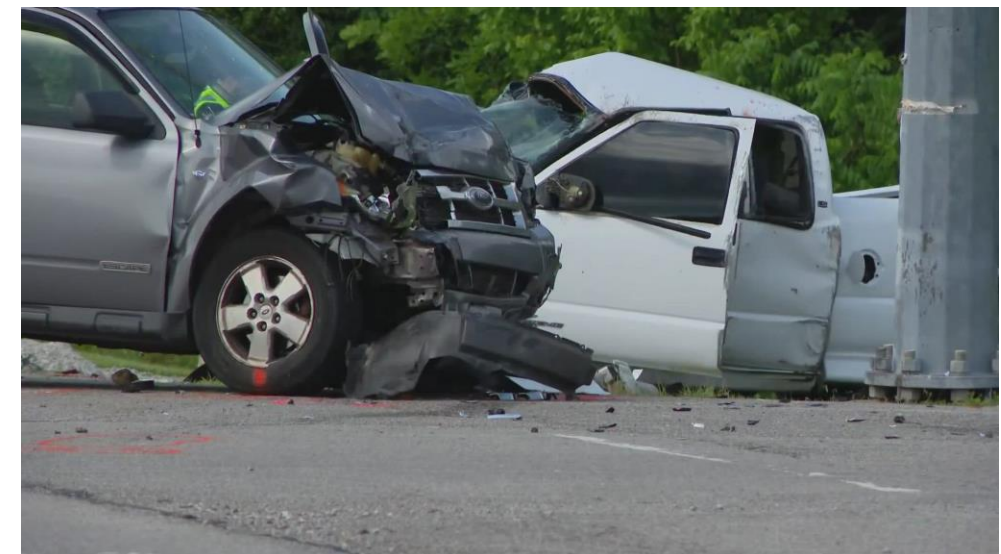
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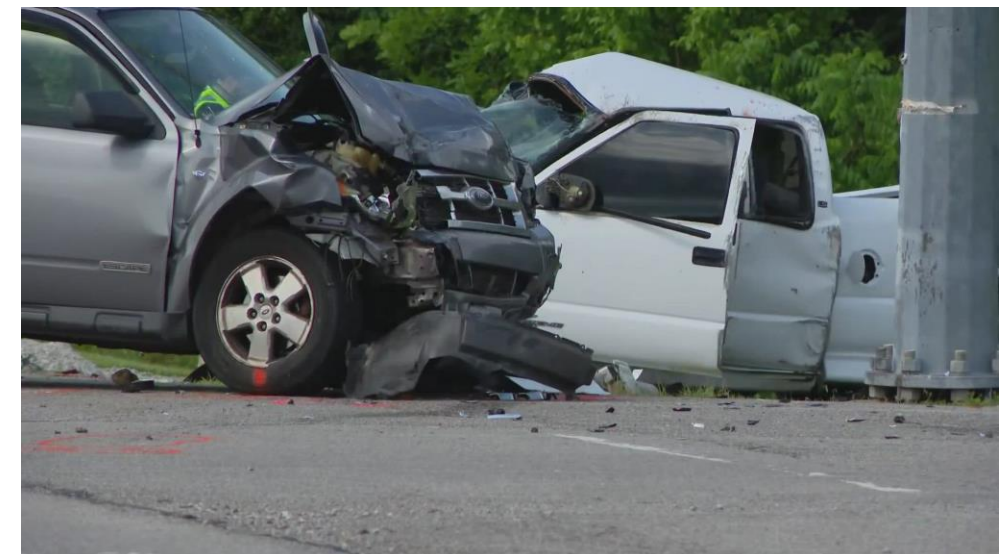
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Fault Lines of Reality: What to Change?

- People undo the same very specific things, the stress points where reality is *slippable* (Hofstadter)
- They tend to change specific aspects:
 - ~ exceptional/non-normative events
 - ~ controllable constraints
 - ~ temporally recent to outcome
 - ~ socially unacceptable
 - ~ focal actor (victim blaming)



Two (of many) Relevant Psychological Theories



- ***Mental Models Theory:***
 - Two models are represented: reality vs. counterfactual alternative → cognitive effort!
 - changes reasoning predictions

Two (of many) Relevant Psychological Theories



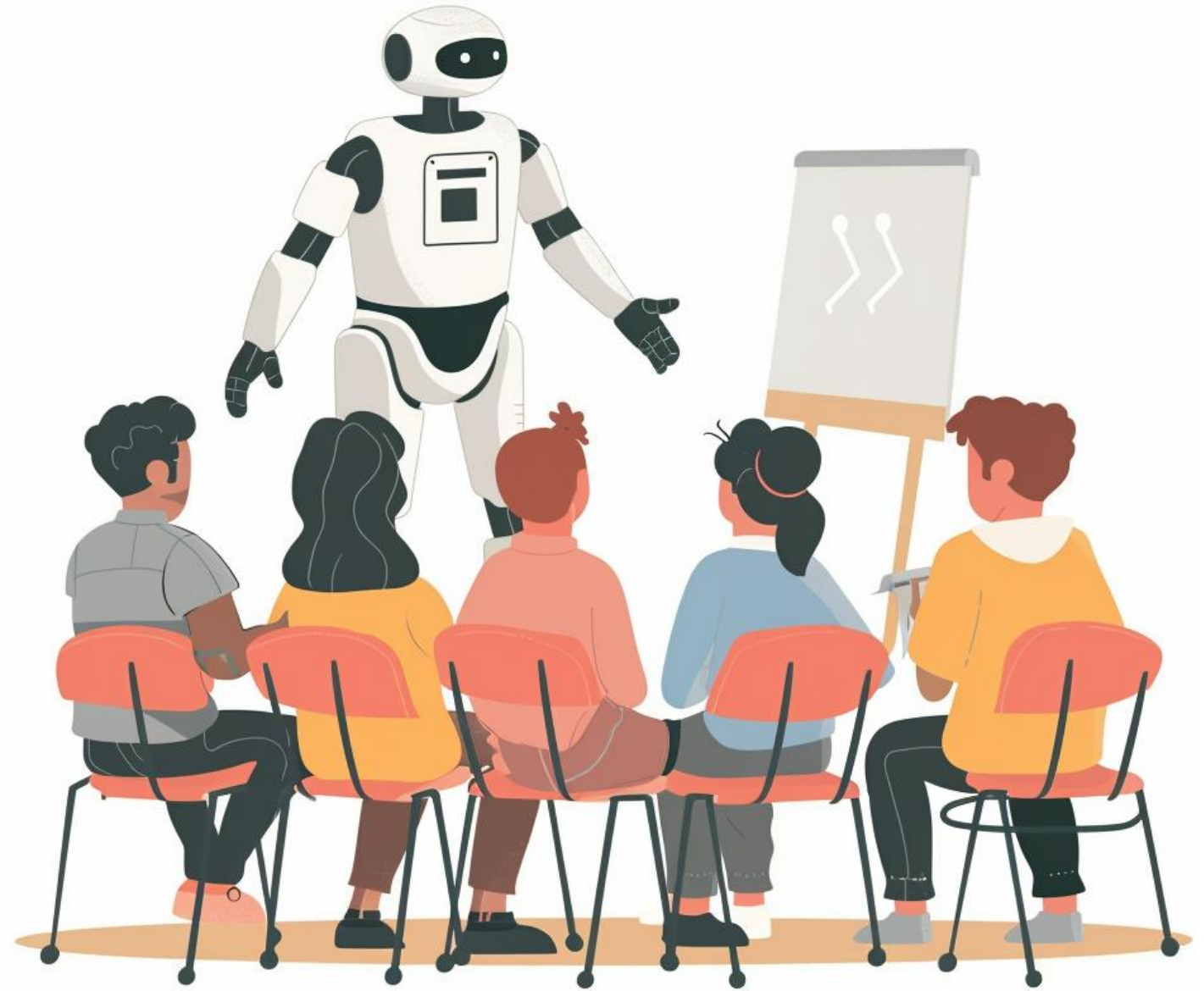
- ***Mental Models Theory:***
 - Two models are represented: reality vs. counterfactual alternative → cognitive effort!
 - changes reasoning predictions
- ***Functional Theory:***
 - High-level account about the consequences for decision making, reasoning, preparation and planning



- They provide “natural” cognitive means to explain outcomes
- CFs are implicated in people’s understanding of causality, so may promote domain knowledge
- They have motivational and emotional impacts
- Lots known about them from psych research
- XAI task (*sort of*) parallels the Normal-Use Task

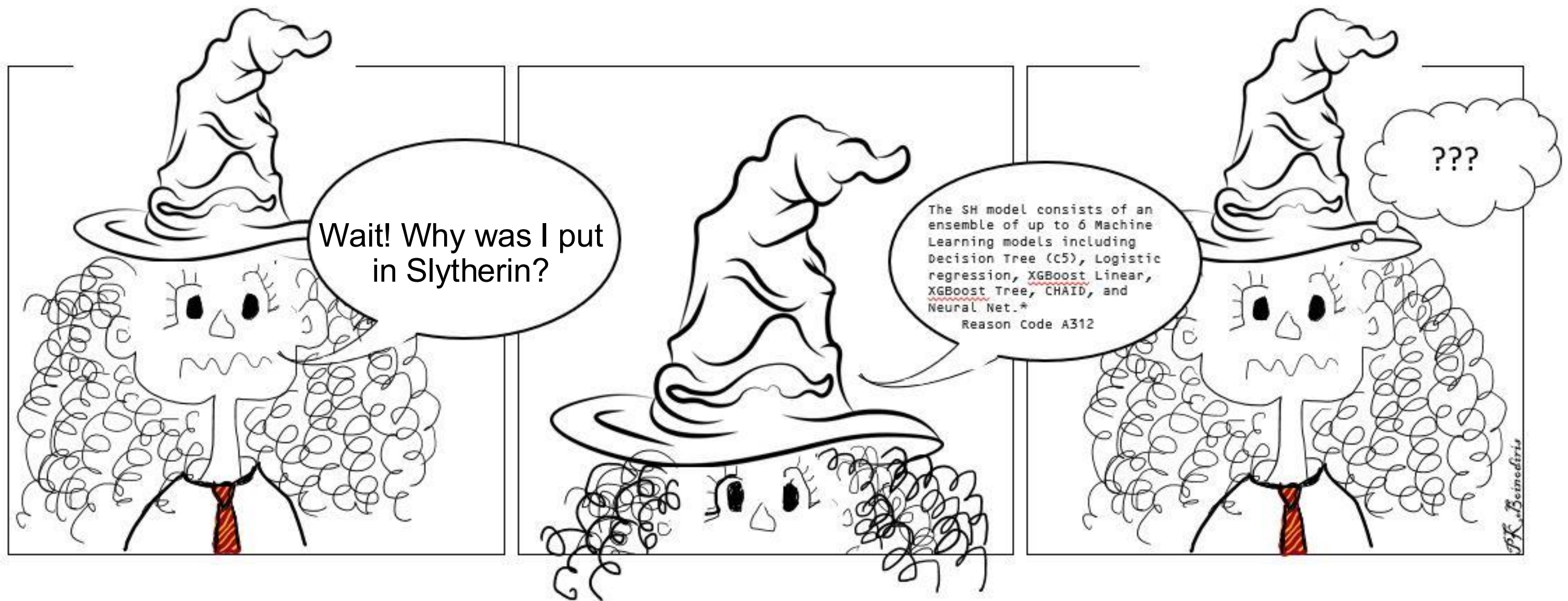


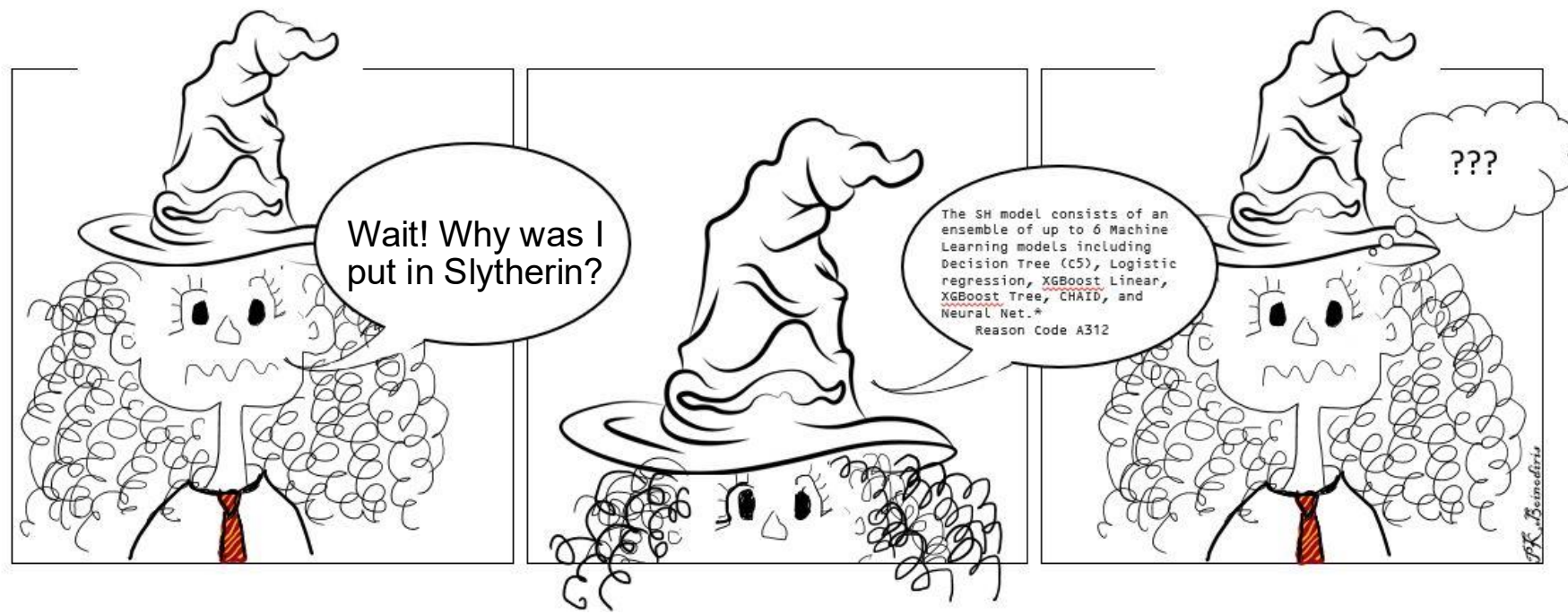
- XAI Task is *not-identical-to* Normal-Use Task !
- Features used/mutated may not be those people typically choose / know about
- Investigating human's mental models / opinions / behavior is complex, but necessary for proper evaluation



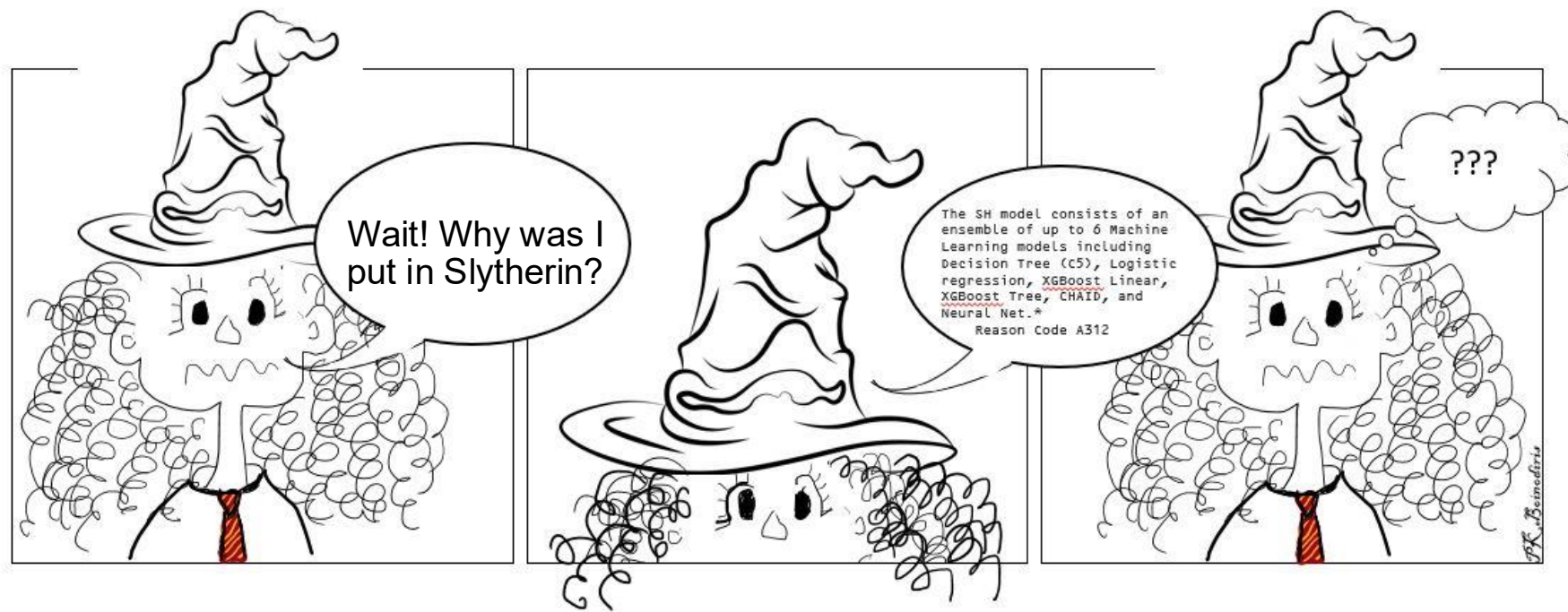
Basics of User Studies

A few first words on the user study “mindset”

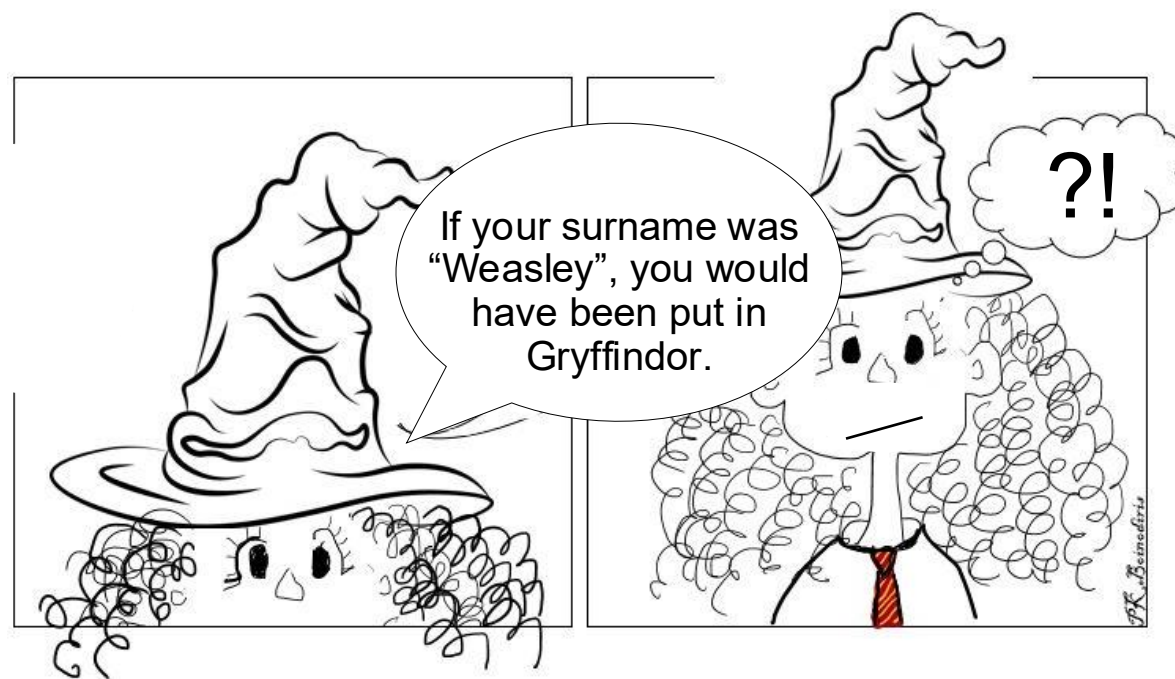


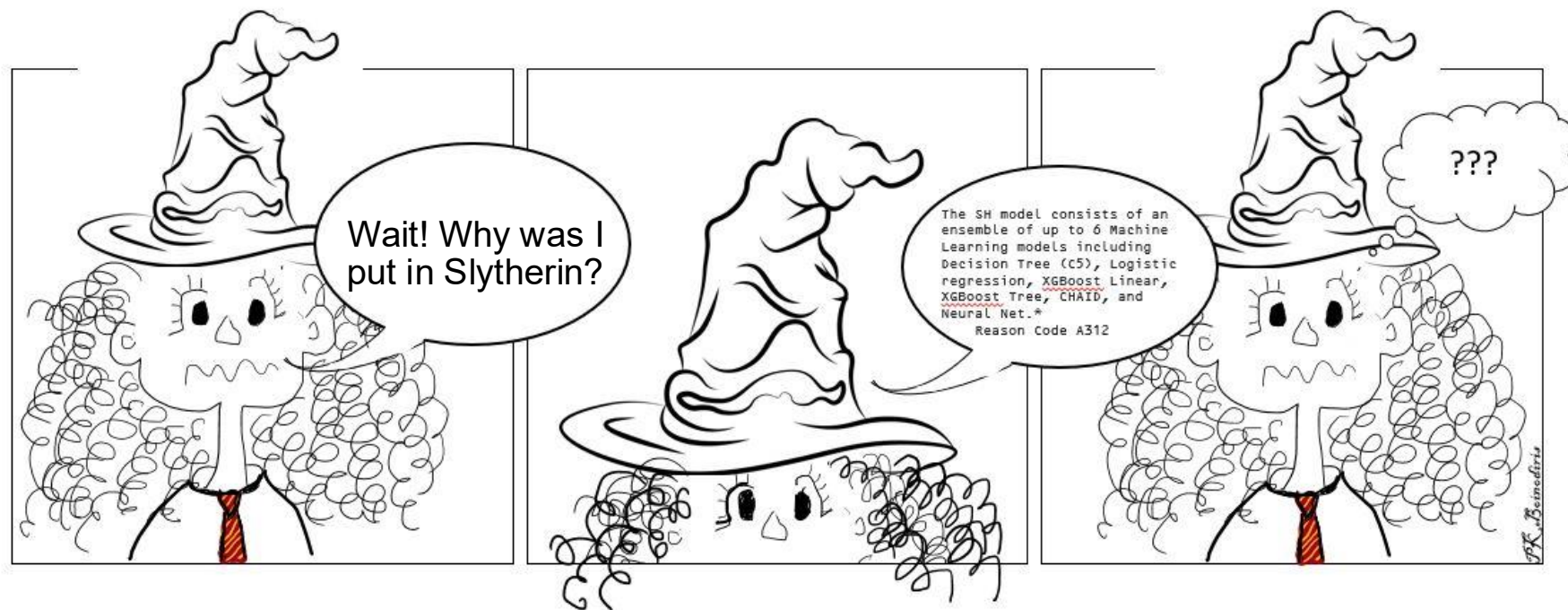


source of original comic:
[Phaedra Boinodiris on Twitter](#)

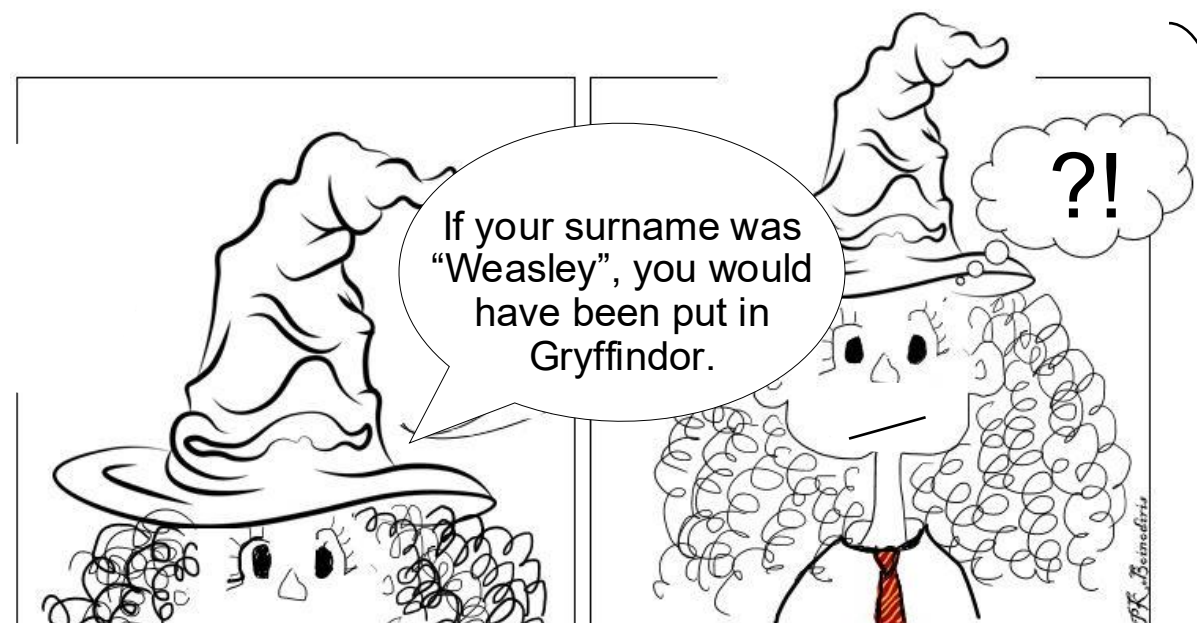


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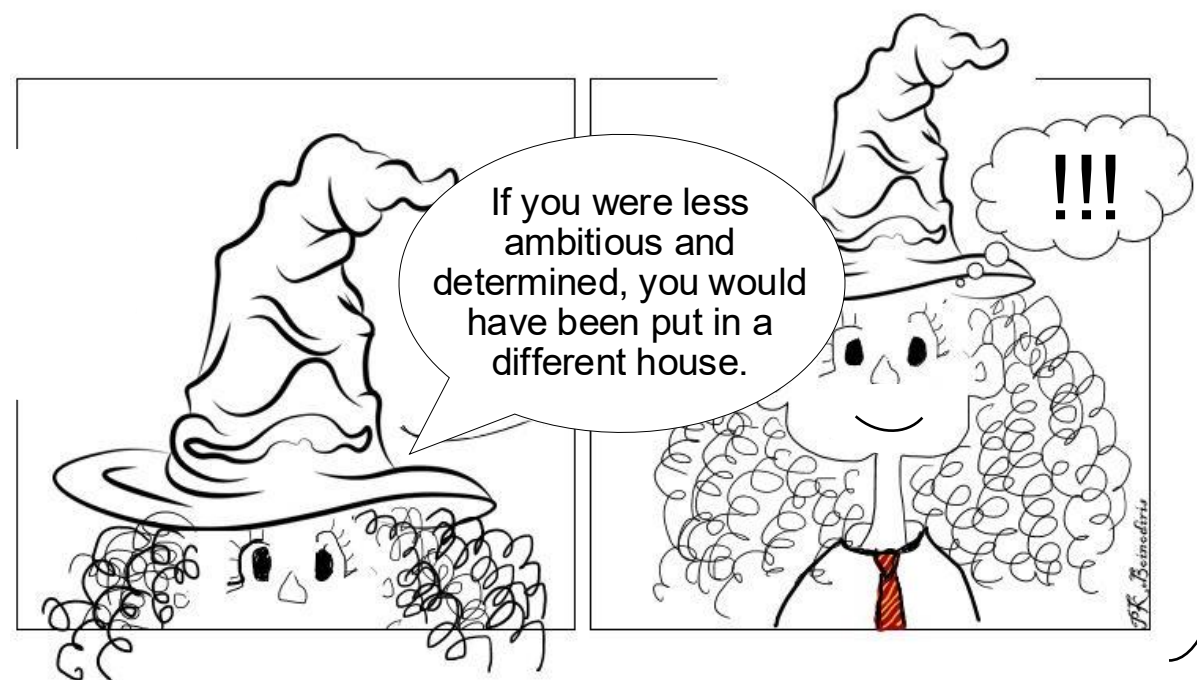




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equally
effective?



equally
satisfying?

equally
trustworthy?

... ask the user!

Designing a User Study

To **what end** do you want to test **what** with **whom** in which **context**?

Designing a User Study

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Purpose/Objective: Examining...



...humans



...models



...human-AI
interaction



...explanations

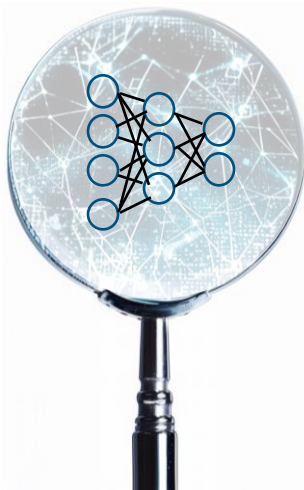
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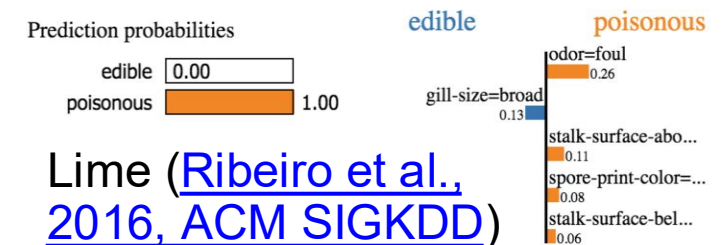


...explanations

Focus - example: compare XAI methods

If the odor had
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VS.



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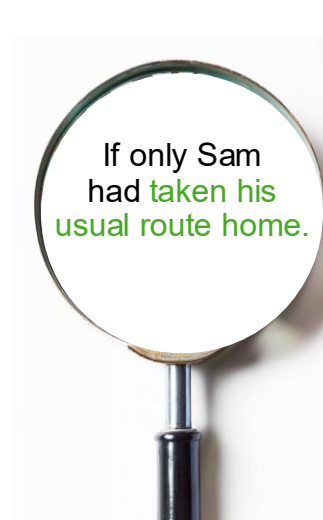
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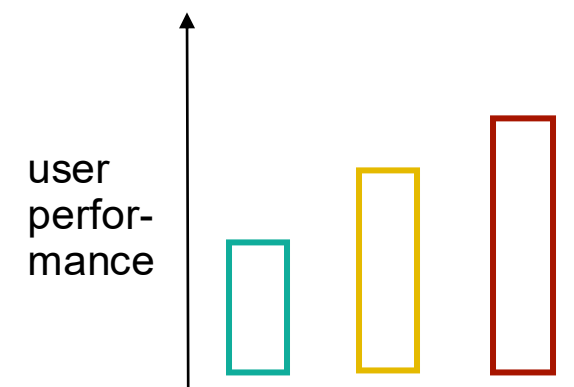
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VS.



Variables – examples: task performance, subjective measures, etc.



Designing a User Study

To what end do you want to test what with whom in which context?

Target User Group:



vs.



Different groups, different challenges:

- sample size
- recruitment
- inclusion / exclusion criteria
- ...

Designing a User Study

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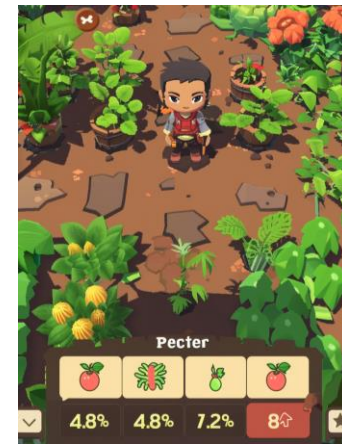
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Application / Environment:



VS.



VS.

„Imagine you're a gardener...“

- User's world knowledge matters!
- User's engagement matters!

Plan for the Day

Time	Topics
1:30 PM	Introduction
	<i>Hello and Introductions!</i>
1:45 PM	From Philosophy of Counterfactuals to XAI and User Studies
	<i>Philosophical & Psychological Fundamentals of Counterfactuals</i>
	<i>Getting Into the User Study Mindset</i>
	<i>Q&A</i>
2:20 PM	Computational Fundamentals of Counterfactuals in AI
	<i>Formalisation</i>
	<i>Current and Future Directions</i>
	<i>Q&A</i>
3:00 PM	COFFEE (15:00h-15:30h)
3:30 PM	Hands-On I: Using a Counterfactual Toolbox
	<i>From a Simple Baseline to Diverse Counterfactuals</i>
	<i>Q&A</i>
4:10 PM	Fundamentals of XAI User Studies
	<i>A Roadmap: From Motivation to Results</i>
	<i>Q&A</i>
4:50 PM	Hands-On II: User Sample Study - Walkthrough and Analysis
	<i>From Raw Data to a Clean Analysis Sample</i>
	<i>Testing & Interpreting Effects of Counterfactual Explanations</i>
	<i>Q&A</i>
5:20 PM	Closing Session, Discussion and Final Q&A
TUTORIAL END	

You Are Here!

WHAT TO REMEMBER

- **Mathematical proximity is one operationalization of closeness; it is not the definition of closeness.**
- **A prediction-flipping counterfactual is not automatically a causal counterfactual.**
- **Algorithmically minimal and psychologically minimal need not be the same thing.**
- **A good counterfactual is always good for someone, for some purpose, in some context.**